
SMAP: A Comprehensive Solution to Plastic Waste Wanaagement - From a Global Perspective

Summary

One of the most significant global emergencies humans are facing today is the overexploitation of plastic usage. While most of the researches on plastic-induced environmental issues is focused on a local or componential aspect, we hereby present our solution to the plastic crisis at a global perspective in the face of the urgent situation.

In this paper, we present a comprehensive analysis at both the macro and micro level with hundreds of research papers and statistics involved. An overall goal for plastic waste is proposed: we show that the growth rate of the plastic industry can work as the key controlling factor when setting the objective. Such the goal can be further distributed to countries and thus establish a viable and practical path for countries. To derive such the objective, some indicators are created to measure relevant properties quantitatively, and mathematical models are established to estimate key values, including the Disposable Limit of Plastic Waste, Total Plastic Production under Maximum annual Capacity of plastic waste, and the environmental safety level of plastic waste.

More specifically, we first predicted the global capacity of handling the plastic waste in 2020 be 69.9 Million tons, by using six main indexes to measure the capability with **Entropy Weight Method** Based on the maximum capacity, we can obtain a target for plastic production for 2020, which is 213 Million tons according to our estimation. **Group decision-making method** and the **variation coefficient method** is used to obtain the relationship between the capacity and the production.

We then distributed the target production goal to different countries, according to their status quo. Five indicators are used to measure a countries status quantitatively, and **PCA** and **spectral clustering** reduce the dimension of the data. The production reduction indicators can be allocated to countries according to the cluster that countries belong to.

We further consider the problem at the time scale. We proposed that **the growth rate of the plastic industry** can be a practical controlling factor. As long as the growth is well controlled, the higher growth trend of its capacity for plastic waste will finally intersect the lower growth curve of industrial production at some point. **Quadratic exponential smoothing time series method** is used for prediction, and we estimate the annual growth rate of garbage disposal by 31.023 million tons/year, and the annual growth rate of total plastic production by 15.652 million tons/year. If further actions were taken by countries that limit the annual growth rate of the plastic industry under 7 Million tons/ year, a safe level for the environment would be reached in just ten years.

Finally, we consider the actions that different countries can take under the context of equality. We turn the global equality problem into a **multi-criterion optimization problem**, which eventually can be converted to a **linear programming (LP) problem**. Hence, a dynamically tunable and practical solution can be given to different countries.

In conclusion, with sufficient information and robust mathematics models, our research provides a viable and scientific solution for dealing with plastic-induced environmental issues, which will have great potential in global sustainable development.

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1 Preface

1.1 Introduction

Since its first invention in the early 20th[1-2], the vast production and application of plastic have dramatically altered people's lives. However, the accompanied growing plastic-induced environmental problem has been worse by day.[3-5]

Some researches have been done in the plastic-induced environmental problem, which includes the source of plastic waste[6-9], the method of handling the waste, and regional policy[13-20], etc. However, most of the current researches are focused on the local or componential aspect of the problem. Only a few studies and reports have been published in analyzing the issue from a global perspective[21][5], but the concentration is still solely presenting the fact. There is an urgent need to find solutions from a global perspective, and the complex factors should be considered if analyzing environmental questions on a worldwide scale.

Hence, we now present our research that proposed a solution in dealing with the plastic crisis our world is currently facing, with the discussion at both the macro and micro levels. By utilizing hundreds of researches and statistics in the past few decades, combined with various algorithms and statistical approaches, our study aims to provide a viable and scientific solution for dealing with plastic-induced environmental issues, which will have great potential in global sustainable development.

1.2 General Assumptions

- The maximum level of plastic waste is measurable, the trend is continuous during a given time span, and is related to factors including the source of waste and the effectiveness of handling the waste, etc.
- The maximum level of plastic waste is defined as the amount of plastic waste that can be handled or processed in a year within the global carrying capacity. Such waste handling practices consist of the artificial practices to dispose waste and the nature-involved process. By using the annual disposable limit of plastic waste as the criterion, further damage to the environment by plastic waste can be eliminated.
- The effectiveness of a particular waste management approach (e.g. mechanical recycling) has a general positive correlation with the technology, economy, policy, and the severity of the environmental problem. (Countries tend to increase the investment, establish positive policy, and foster the technological advances when encountering severe environmental problem, which allows more plastic waste to be managed appropriately)

1.3 Notations

We list the symbols and notations used in this paper in Table 1

Table 1: Notations

Symbol	Definition
DLPW	Disposable Limit of Plastic Waste
SMAP	Synthesis Multicomponent Analysis for Plastic waste
ECPW	Environmental Capacity of Plastic Waste
Technique _(i)	Maximal amount of plastic waste that can be handled by i^{th} method
Technique _(j) Technique _(k)	Maximal amount of plastic waste that can be handled by k^{th} method by different countries, $j=1, 2, 3$ denotes high-income country, medium-income country, and low-income country, respectively
$C_{(i)}$	Capability index for handling plastic waste by i^{th} method
$C_{(j)}$ $C_{(k)}$	Capability index for handling plastic waste by i^{th} method by different countries, $j=1, 2, 3$ denotes high-income country, medium-income country and low-income country, respectively.
$M_{(i)}$	The weight for high income country, middle income country and low income country

2 Task 1: Estimate the Disposable Limit of Plastic Waste

2.1 Model: SMAP

We propose a SMAP to estimate the disposable limit of plastic waste (DLPW), which takes the total amount of Artificial disposal and natural-involved process in handling the plastic waste into account. SMAP considers two categories of handling the plastic waste: recycling and appropriate disposal, based on factors including the proportion of plastics with different physical and chemical properties used in industry and commerce, level of harm to nature, the policy, and degree of attention, etc. The capacity of recycling can be measured by the total amount of mechanical recycling, energy regeneration, material recovery, chemical recycling, and biodegradation, while the disposal amount can be estimated by the capacity of incineration and landfill under ECPW.

$$DLPW = \sum_{\text{Recycling}} \text{Technique}_{(i)} + \sum_{\text{Proper Disposal}} \text{Technique}_{(j)} \quad (1)$$

$$\text{s.t. } \text{Technique}_{(j)} \leq EPCW \quad (2)$$

It is worth noticing that for each technology, for example, the mechanical recovery, if we simply estimate current or future annual throughput based on past data, the result will be inaccurate due to the following constraint.

- Indicators such as current or future countries' technology development, industrial development, policies, government budget, and severity of the environmental problem,

etc., alter annually. In a long-time period, the abovementioned indicator shows a dynamic nonnegligible change.

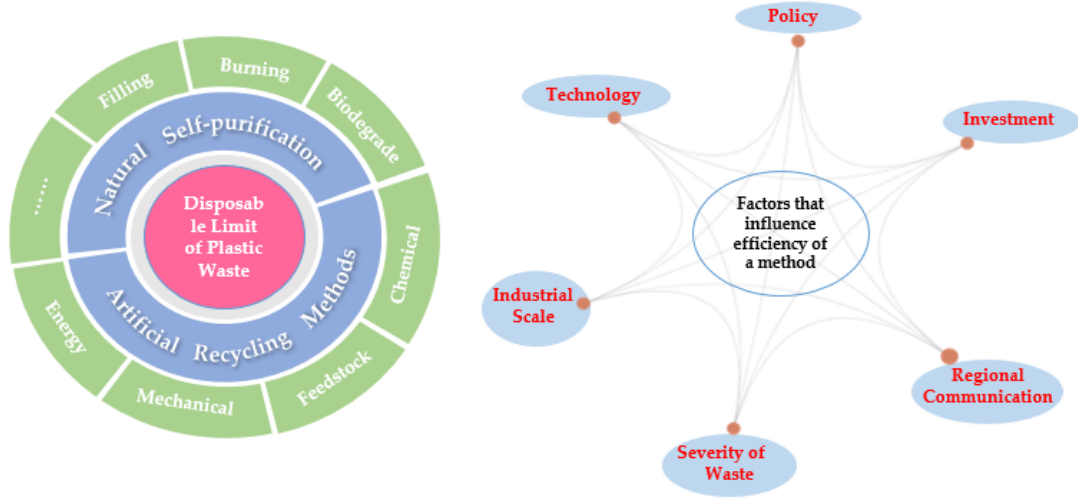


Figure 1: The scheme of our SMAP model

- The abovementioned indicator applies to every country, but the influences of indicated factors differ based on the GDP level of countries.

2.2 Comparative Analysis

Consider the abovementioned constraint, **comparative Analysis** is utilized to **enhance the model's robustness and generalization**.

Step 1 First, we choose a specific time point as our start point for analysis (in our case, we select the year 2010). The maximal amount of plastic waste that can be handled by each method ($\text{Technique}_{(k)}^{(j)}, j = 1, 2, 3$) is used for the estimating future trend. Next, we use **the Entropy Weight Method**. We measured every indicator and given each indicator a weight w_i based on its variability. $C_{(k)}^{(j)} (j = 1, 2, 3)$ can be calculated for three types of countries.

Step 2 Considering a given time in the future (e.g., the year of 2020), the weight w_i for each indicator is assigned with corresponding values. By applying the EWM, $\text{New_}C_{(k)}^{(j)} (j = 1, 2, 3)$ for handling plastic waste, can be calculated. The capability for the corresponding technic can be then calculated.

$$\text{New_Technique}_{(k)}^{(j)} = \text{Technique}_{(k)}^{(j)} \times \frac{\text{Newtext}C_{(k)}^{(j)}}{C_{(k)}^{(j)}} \quad (3)$$

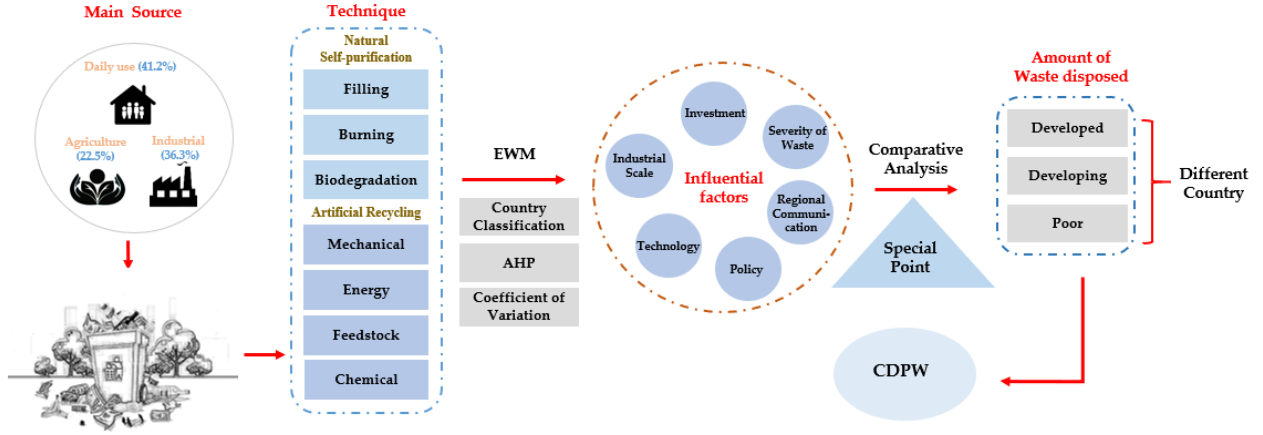


Figure 2: The framework of SMAP model

$\text{New_Technique}_{(k)}$ is the total amount of plastic waste that can be handled globally.

$$\text{New_Technique}_{(k)} = \sum_{j=1}^3 \text{New_Technique}_{(k)}^{(j)} \quad (4)$$

Step 3 For Entropy Weight Method, as long as the same latitude scalar for each indicator can be applied, the estimation can be done. The value of indicators can be retrieved from multiple sources.

2.3 EWM Process

Given 6 Indicators, X_1 : technology development, X_2 : policy, X_3 : government investment, X_4 : severity of the environmental problem, X_5 : industry scale, X_6 : regional communication. Where $X_1 = \{x_1, x_2, x_3\}$ represent three kinds of countries based on GDP. The value of indicators is normalized to fit in the 0-1 range.

For all five indicators we proposed, including current or future countries' technology development, industrial development, policies, government budget, and severity of the environmental problem, have a positive correlation to CI. Hence, following system of equations can be derived:

$$\begin{cases} y_{ij} = \frac{X_{ij} - \min(X_i)}{\max(X_i) - \min(X_i)} \\ \text{(or) } y_{ij} = \frac{\max(X_i) - X_{ij}}{\max(X_i) - \min(X_i)} \end{cases} \quad j = 1, 2, 3 \quad (5)$$

After normalization, we can use $y_{i,j}$ replace of $X_{i,j}$ to denote C_i for a specific country. Hence

we get:

$$p_{ij} = \frac{y_{ij}}{\sum_{j=1}^3 y_{ij}} \quad (6)$$

The information entropy of each evaluation index can be calculated, and the information entropy of each evaluation index can be obtained base on the concept of information and concept of entropy in information theory

$$E_i = -\ln(n)^{-1} \sum_{j=1}^3 p_{ij} \ln(p_{ij}) \quad (7)$$

$$w_i = \frac{1 - E_i}{1 - \sum_i E_i} \quad i = 1, 2, 3, 4, 5 \quad (8)$$

Finally we can get the capability index for handling plastic waste by $Technique_{(k)}$

$$C_{(k)}^{(j)} = \sum_{i=1}^5 w_i E_i \quad j = 1, 2, 3 \quad (9)$$

2.4 DLPW of 2020

By applying EWM method, we can predict the weights of indicators for the three types of countries in 2010 and

Table 2: The weights of indicators for the three types of countries in 2010 and 2020

Managed Treatment Level	Index	Weight						
		Country Type	High-income		Middle-income		Low-income	
		Year	2010	2020	2010	2020	2010	2020
	Existing Technology		0.31	0.38	0.19	0.25	0.17	0.20
	Policy		0.22	0.24	0.25	0.27	0.39	0.36
	Capital Investment		0.11	0.12	0.08	0.11	0.03	0.09
	Severity		0.21	0.09	0.32	0.19	0.25	0.21
	Industrial Scale		0.04	0.05	0.05	0.05	0.12	0.07
	Regional Communication		0.11	0.12	0.11	0.13	0.04	0.07
overall merit / weighted sum		0.71	0.86	0.41	0.52	0.25	0.31	

From the SMAP model, we can predict the disposable Limit of Plastic Waste(DLPW), which is the Maximal amount of plastic waste that can be handled by three types of countries by 2020

The global disposable limit of plastic waste (DLPW) can be derived from statistics in Table 2 by calculating the sum of each column. Therefore, we hereby offer our prediction for the global disposable limit of plastic waste (DLPW) in 2020, which is **69.9MT**.

Table 3: The results from the SMAP model

Technique		Harmless Treatment Capacity /million ton per year						
		Country Type	High-income		Middle-income		Low-income	
		Year	2010	2020	2010	2020	2010	2020
Recycling	Mechanical		4.6	5.6	10.3	13.0	0.9	1.1
	Energy recovery		5.8	7.0	13.9	17.5	2.1	2.5
	Feedstocck		1.2	1.5	2.8	3.5	0.3	0.4
	Chmeical		0.6	0.7	2.4	3.0	0.2	0.2
	Biodegradation		0.1	0.1	0.5	0.6	0.1	0.1
Disposal	Filling		0.3	0.4	2.3	2.9	0.2	0.2
	Burning		2.7	3.3	3.7	4.7	1.4	1.7
Total			15.3	18.5	35.9	45.2	5.2	6.1

3 Task 2: Allocate the target goal to countries

3.1 From DLPW to TPPMC

In problem 1, we have discussed the estimation model of the Maximum annual Capacity of plastic waste. However, such indicator measured and estimated by waste management approaches is not intuitive for the evaluation of environmental safety level. Therefore, we introduce the indicator **Total Plastic Production under Maximum annual Capacity of plastic waste, or TPPMC**, as the minimum environmental safety level, which can be intervention by human directly and effectively. It is obvious that the disposable limit of plastic waste, or DLPW and TPPMC are positively related, and their conversion efficiency is affected by indicators including technology, economy, policy, crisis severity, interregional connection, etc.

The model based on **group decision-making method** to estimate the individual impact of different indicators on the conversion efficiency of TPPMC to DLPW, and **the comprehensive impact of all indicators** are evaluated based on that method.

- 1) In order to achieve a reasonable result, six intermediate indicators from previous section are introduced: technology development, policy, government investment, severity of the environmental problem, industry scale, and interregional communication. Therefore, for $CE_{(i)}$, the maximum value of i is 6. Then introduce the evaluation E of $CE_{(i)}$ by experts in different fields. In our case, information from is considered, which covers four fields: conomy, politics, environmental protection and culture.
- 2) We use **CVM (coefficient of variation method)** to give dimensionless weight to the evaluation index C_v , $C_v \in (0, 1)$ in different fields. Suppose that evaluation from n different experts are collected in a certain field to evaluate $CE_{(i)}$ (for example, $i = 1$, which means

capital investment) N), hence:

$$\text{where } \begin{cases} C_v = \frac{\sigma}{\mu} \times 100\% \\ \mu = \frac{1}{n} \sum_{i=1}^n E_i \\ \sigma = \sqrt{\frac{1}{(n-1)} \sum_{i=1}^n (E_i - \mu)^2} \end{cases} \quad (10)$$

- 3) Let $C_v^{I,j}$ denotes The overall evaluation of the experts in the on the i th indicator, the following 4×6 evaluation matrix can be derived

$$E = \begin{bmatrix} C_v^{(1,1)} & \dots & C_v^{(1,6)} \\ \vdots & \ddots & \vdots \\ C_v^{(4,1)} & \dots & C_v^{(4,6)} \end{bmatrix} \quad (11)$$

- 4) Hence, the reverse process to calculate TPPMC from DLPW is

$$\frac{\text{DLPW}}{\text{TPPMC}} = \frac{\|EE^T\|^2}{\sum_{i=1}^4 \lambda_i}, \quad \text{diag}(EE^T) = \begin{bmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_4 \end{bmatrix} \quad (12)$$

- 5) Plug in the number, in our case, the DLPW of 2020 is **69.9MT**, so that the TPPMC of 2020 can be calculated with the value of **213MT**

3.2 Clustering indexes

According to the result derived in previous step, the safety level of total amount of the global plastic production is **213MT**, while the predicted current level of plastic production is **437MT**. The gap between the safety capacity and the current level indicates that the at least **48.74%** production should be eliminated. The reduction in production is ought to be shared by different countries.

- Economics

discussed in previous section, the development of economics plays a critical role in the amount of plastic waste that can be handled by a specific country. We use the widely used indicator, GDP per capita, to denote the economic development of a country. A normalization operation is performed before the indicator is used for clustering. Let x_i denotes the GDP per capita of a given country, we get

$$\hat{x}_i = \frac{x_i - \min}{\max - \min} \quad (13)$$

- Policy

We have shown that there is a relationship between the government attitude and the amount of waste that can be handled by a specific country. We illustrate this point

further in this section. More specifically, following policies applies to different countries in the world: Total or partial ban □ Economic instruments □ Combination

According to , given the weight to Total or partial ban as 0.75, economic instruments as 0.5, combination as 0.5

- Technology

Higher level of technological development will also greatly enhance the waste-handling capability, which we have discussed in the previous section. In this section, we quantitatively apply the ratio of expenditure on scientific research to GDP to measure the importance of technology that a country has attached to. If a country emphasis on development of high technology, not only can the capacity of waste-handling be increased, but also more likely to develop and utilize new environmentally-friendly material to replace plastic.

- Amount of plastic waste

The total amount of plastic waste reflects a country's current environmental situation. We utilize source from to retrieve such data of each country, by including data for waste parings, and scrap of polyethylene (3915.10), polystyrene (3915.20), polyvinyl chloride (3915.30), and other plastic n.e.s. (3915.90) for years 1988 -2016. Similarly, we have normalized the amount of plastic waste considering the size of countries. Hence, the indicator is the output of plastic waste is

$$y_i = \frac{DLPW}{GDP} \quad (14)$$

- The amount of exported plastic waste

Exporting waste is an important way of handling plastic waste for some countries, especially some developed countries, we use the notation export to denote the amount of exported plastic waste, import to denote the amount of production of the plastic waste. Hence,

$$index = \frac{export}{production} \quad (15)$$

- Amount of plastic waste that is harmless treated

Some countries have data on the amount of waste that is treated in a harmless manner, Such data is used for auxiliary classification, because they are not easy to obtain, with many countries have these statistics unreported.

3.3 Dimension reduction

Since some of the abovementioned indicators are associated, and the high-dimensional data is not conducive to visualization, **PCA algorithm** is applied before clustering. Dimensionality reduction processing is carried out for the indicators 1-5, and the result is a three dimension vector.

3.4 Clustering

A widely used clustering method is K-Means approach, which divide sample into k^{th} Voronoi area. However, after implementing K-means clustering, the result cluster works unwell, due to the constraint that K-means can only handle convex set, while our dataset we use, which is visualized in Figure 3 , possesses a non-convex property. Therefore, **spectral clustering**

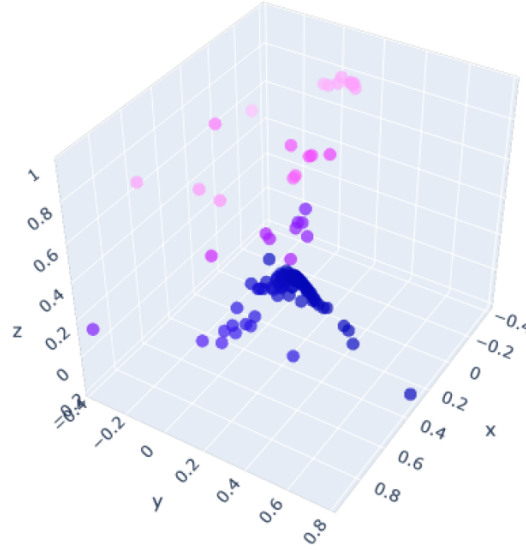


Figure 3: Data distribution

algorithm is applied in our model, which utilizes **the spectrum (eigenvalues) of Laplace matrix** based on the measurement of similarity to cluster the data that possess non-convex property. In our implementation, simulation matrix is established using **Gaussian kernel function (Radical Basis Function)**

$$K(x, z) = \exp \left(-\gamma \|x - z\|^2 \right) \quad (16)$$

After the cross-validation with the generated matrix with the raw data, we let $\gamma=0.1$, the number of cluster=5. Thus we can get the reduction of plastic production in different clusters estimated by our model.

Table 4: Classified results

Classification	Indexs
Cluster 1	17.90%
Cluster 2	13.10%
Cluster 3	41.30%
Cluster 4	26.30%
Cluster 5	1.40%

The following table shows the more details of 51 countries that have been classified into 5 clusters.

Table 5: Different clusters for 51 countries (full version refer to Supplemental Material)

Cluster1	Cluster2	Cluster3	Cluster4	Cluster5
Oman	Netherlands	China	United States	Germany
Mexico	Israel	Nigeria	Australia	Japan
Bulgaria	France	India	Sweden	
Guinea-Bissau	Italy	Pakistan	Canada	
Vanuatu	Spain	Sudan	United Kingdom	
Curacao	Panama	Vietnam	Singapore	
Ukraine	Romania	Indonesia	New Zealand	
Djibouti	Turkey	South Africa	United Arab Emirates	
Honduras	Kenya	Russia	Ireland	
Ghana	Albania		Norway	
Solomon Islands	Morocco		Greece	
Nicaragua	Chile			

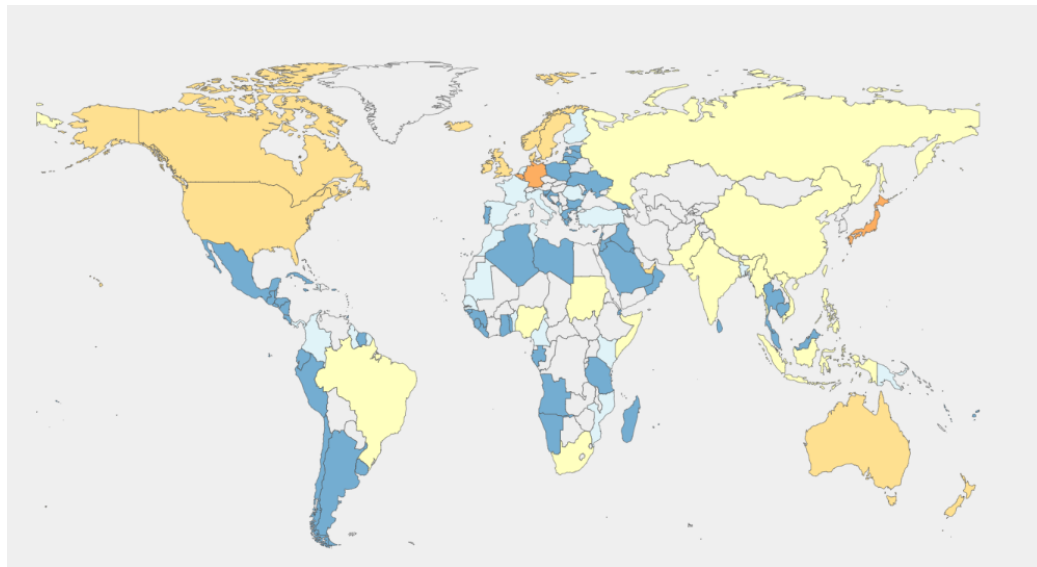


Figure 4: Clustering results, different color denotes different type

3.5 Weighted model

The amount is derived from the following analysis:

Let the annual production amount for a given cluster be x_i , the quantity of reduction required be y_i , n denotes the number of that cluster, and m denotes to total number of countries, in our case, $m=51$. Hence, the reduction of plastic production that is recommended for

countries in different clusters can be derived as:

$$y_i = \frac{\sum_{i=1}^n x_i}{\sum_{j=1}^m x_j} \quad (17)$$

- **Cluster 1**

The countries in this cluster are mainly consists of developing countries, and can be further divided into countries with higher-income and lower-income. Their common feature is that plastic waste production and waste exports is below average, have with no legislation restricting plastic production.

It can be calculated the reduction of plastic production that is recommended for countries with higher income in cluster 1 is 13.6%, that is, 30.46mt of reduction per year. These countries do not have strict national legislation restricting the production of plastics, while some other countries with similar economic development have already done so. Therefore, the reduction suggested is under the capability of these countries. Some specific viable path including set laws that restrict plastic packaging, which currently accounts for 36%, of total plastic waste. Research and import plastic alternatives, or utilize ways to make plastic waste less harmful to the environment is another viable solution.

Meanwhile, calculated reduction of plastic production that is recommended for countries with lower income in cluster 1 is 4.3% , that is, 9.6mt of reduction per year. Low production and lower level of economics and industrial development suggests that limiting the household plastic through legislation and choose lower plastic-consuming industry during industrialization may be a viable path.

- **Cluster 2**

Countries in cluster 2 mostly consist of developed countries, such as France and Italy, and they have national restrictions and policy on plastics and produce less plastic waste. However, advanced technological approach can be applied, for example, the development of materials that replace plastic. The resultant calculated reduction of plastic production that is recommended for countries in cluster 2 is 13.1% , that is, 29.34mt of reduction per year

- **Cluster 3**

Cluster 3 countries, such as China, Russia and other developing countries, produce a large amount of plastic per capita. Although there is some national legislation and policies, the economic development relies too much on plastic-related industry and the production of plastic-related products, which makes the reduction challenging. However, due to the existing large plastic production amount, there is actually a huge space for reduction. The resultant calculated reduction of plastic production that is recommended for countries in cluster 2 is 41.3% , that is, 91.8mt of reduction per year. A suggested viable path includes adopting stricter plastic restrictions, such as reducing the production or even bans the usage plastic bags. More investment should be made in the research and development of plastic substitutes and recycling methods.

- **Cluster 4**

Cluster 4 countries, such as the United States and Canada and other developed countries, have well developed economies, but with no or less national legislation or policy on restricting plastic waste. Therefore, by limiting the consumption of plastics in the industrial and civil sectors at the legislative level, with the assistance of technology and the development of plastic-recycling industries, the reduction of plastic production that is recommended for countries in cluster 4 can reach a level to 26.3% , that is, 58.91mt.

- **Cluster 5**

The countries in Cluster 5 are Germany, Japan and Belgium. the reduction of plastic production that is recommended for countries in cluster 4 can reach a level to 1.4% , that is, 3.13mt, due to the existing low level of plastic waste and high recycling rate in these countries. Currently, most part of the plastic waste generated by these countries is exported. The target can be achieved by further utilize environmental-friendly plastic alternatives and the help of technology in handling plastic wastes.

4 Task 3: Reaching the Target via the Annual Growth Rate of Plastic Industry

4.1 Analysis on the hard but imperative goal

4.1.1 Factors considered

In 2020, According to the result derived in the previous question, the safety level of the total amount of global plastic production is 213MT, while the predicted current level of plastic production is 437MT, among which the plastic industry is playing influential part. Industrial, agricultural and commercial demand calls for the development of plastic industry, and it is therefore not possible to half the current production of plastic in a single year. In order to achieve the desired safety level, new approach has to be put into consideration. We hereby presents a way to deal with that problem. First, consider the following factors

- More attention has been paid to eco-friendly plastics under the transformation and upgrading of plastic industry. Scientific and technological advances allows alternatives of plastic a possible choice in many fields.
- The progress of science and technology allows mechanical recovery and chemical recovery process have higher efficiency. In the future, biological recovery will also become an important part in plastic management. The abovementioned trend shows the potential increase of the capacity of plastic waste handling
- The environmental protection organizations and the governments of different countries coordinate and cooperate to introduce policies and legitimations to limit the development of non-biodegradable plastics

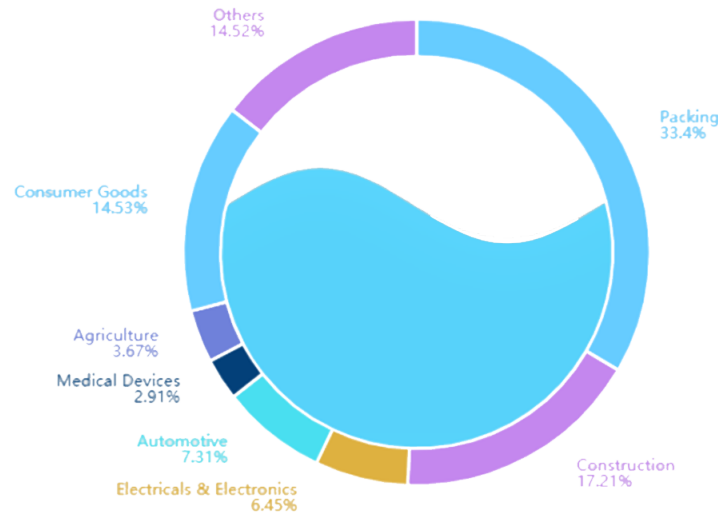


Figure 5: The source of plastic waste

- The combination of plastics and agricultural industries (such as plastic film technology and insulation film technology) has greatly increased the output of crops and reduced the use of water, which is an important factor that cannot be ignored in the development of economy

4.1.2 Strengths that lie in the future

Therefore, **annual growth rate of plastic industry** can be set as the goal. although the current plastic production is still beyond environmental safety level, but as long as the there are science advances, more positive policy, people's rising awareness of environmental protection, the amount of plastic waste that can be handled will grow continuously. Moreover, if the plastic industry growth is lower than the growth rate of maximum handleable waste amount, the curves of expected plastic production and the expected environmental safety level of production will eventually intersect, indicating that the plastic production is at an appropriate level.

The abovementioned approach has the following advantages:

- After the year of the intersection point, annual plastic production will remain steadily below the safety level for environment for a long time
- Although the growth rate of the plastics industry has decreased, the industrial output value is still growing at a low pace, which is still beneficial to the development of industry, commerce and agriculture
- In the long-term, the growth rate of the plastics industry can be adjusted, for example, increased, according to the demand (still maintained at the level below the environmental safety limit), which increases the flexibility of future subjective adjustment

- Sacrifice part of the contemporary interests for long-term development in the future complies with the principle of sustainable development

4.2 Modeling result: Increased by 7MT/year

In order to derive an effective annual growth rate of the plastics industry, we first consider the long-term development curve of the amount of plastic waste that can be properly managed with following steps

First, the weights of the three types of countries are weighted and summed to obtain the comprehensive weight M

$$M = 0.35M_{(1)} + 0.5M_{(2)} + 0.15M_{(3)} \quad (18)$$

Based on the influence weights of six factors of three types of countries from 2016 to 2019, the weight prediction model is established by the second exponential smoothing method. We can derive the following system of equations:

$$\begin{cases} S_t^{(1)} = \alpha y_t + (1 - \alpha)S_{t-1}^{(1)} \\ S_t^{(2)} = \alpha S_t^{(1)} + (1 - \alpha)S_{t-1}^{(2)} \end{cases} \quad (19)$$

Where $S_t^{(1)}$ is the smoothed main indicator, $S_t^{(2)}$ is the smoothed second indicator. In order to make the annual growth rate of the plastic industry to a static quantity, we use the linear model to express the trend of parameters:

$$\begin{aligned} M_{t+m} &= a_t + b_t m, m = 1, 2 \dots \\ \begin{cases} a_t = 2S_t^{(1)} - S_t^{(2)} \\ b_t = \frac{\alpha}{1-\alpha} (S_t^{(1)} - S_t^{(2)}) \end{cases} \end{aligned} \quad (20)$$

M_{t+m} represents the predicted weight of the index, and we can determine the value in the following Table 6

Table 6: Comprehensive weight predicted by model

Index	Comprehensive Weight									
	Year	2016	2017	2018	2019	2020	2021	2022	2023	2024
Technology Development		0.26	0.28	0.32	0.37	0.4012	0.4324	0.4636	0.4948	0.5261
Policy		0.27	0.28	0.27	0.29	0.3573	0.4246	0.4919	0.5592	0.6265
Capital Investment		0.10	0.12	0.14	0.14	0.1414	0.1428	0.1442	0.1456	0.1470
Severity		0.21	0.19	0.19	0.17	0.1564	0.1428	0.1291	0.1155	0.1019
Industrial Scale		0.06	0.08	0.10	0.11	0.1202	0.1304	0.1406	0.1508	0.1611
Regional Communication		0.11	0.13	0.17	0.21	0.2402	0.2704	0.3006	0.3308	0.3611

Predicted by model

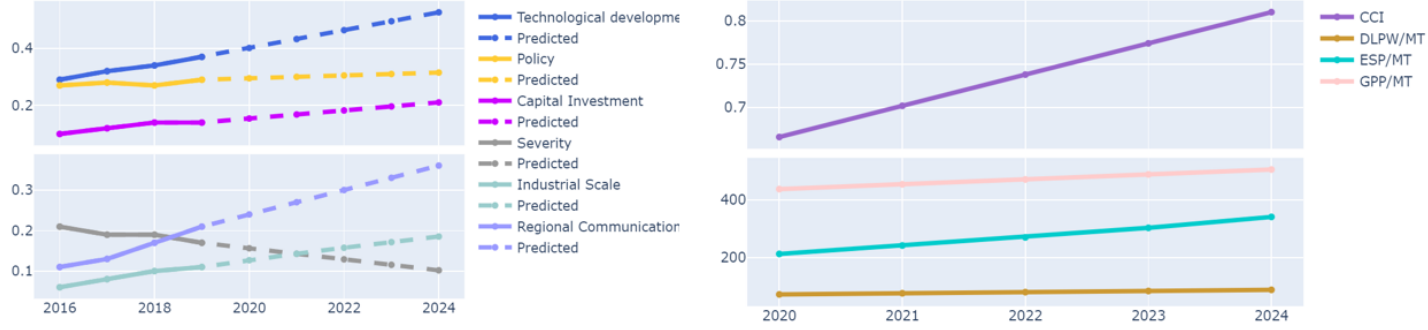


Figure 6: Prediction of weight

Based on DLPW of 66.5mt and CCI (Comprehensive capacity index) of 0.6075 in 2019

$$DLPW_{(year)} = 66.5 \times \frac{CCI_{(year)}}{0.6075} \quad (21)$$

It is complicated to apply entropy weight method to every CCI. Hence, we consider the correlation between two indicators (covariance coefficient and expected value are obtained according to the weight data of 2016-2019). i, j denotes the covariance of i^th and j^th correlation. e.g. $i=2$ denote "Policy" in following table.

$$\rho_{ij} = \frac{Cov(i, j)}{\sqrt{D(i)D(j)}}, i = 1, 2, \dots \quad j = 1, 2, \dots \quad (22)$$

$$\text{where } \begin{cases} Cov(i, j) = E[ij] - E[i]E[j] \\ Cov(i, j) \text{ if } i = j \end{cases} \quad (23)$$

$$E = \begin{bmatrix} \rho_{11} & \cdots & \rho_{16} \\ \vdots & \ddots & \vdots \\ \rho_{61} & \cdots & \rho_{66} \end{bmatrix} \quad (24)$$

A 6×1 matrix is obtained through PCA dimensionality reduction

$$M = [0.7321, 0.6556, 0.2031, 0.3079, 0.2611, 0.2817]^T \quad (25)$$

The weight of 6 indicators in a given year is

$$W_{(year)} = [w_1, w_2, w_3, w_4, w_5] \quad (26)$$

$$CCI_{(year)} = W_{(year)}M \quad (27)$$

Based on this model, Considering both Environmental Safety Production ESP (Environmental Safety Production) and Global Plastic Production GPP (Global Plastic Production). DLPW and CCI for 2020-2024 are obtained as follows.

Applying **least squared method**, The estimated annual growth rates of ESP and GPP are 31.023mt /year and 15.652MT/year, respectively. Considering the different bases level and the impact of plastic production on economic development, we hereby give **our suggested goal that the annual growth rate of plastic industry is set at 7MT/year, then it will only take about 10 years for global plastic production to reach the environmental safety level.**

Table 7: DLPW and CCI for 2020-2024

Year	2020	2021	2022	2023	2024
CCI	0.704	0.778	0.852	0.927	1.001
DLPW/MT	77.052	85.178	93.305	101.431	109.557
ESP/MT	213.021	242.132	270.210	303.211	341.112
GPP/MT	437.353	453.929	470.762	488.210	505.137

4.3 Effect

4.3.1 Plastics industry

On the one hand, current annual growth rate of the plastic industry is more than 8%. If we take its huge base into account, the annual growth rate will be at least 32MT. If that value is reduced to 7MT/year, the revenue of plastic industry will drop by at least a quarter. On the other hand, due to the rigid and incremental demand of construction industry, agriculture and other plastics-needed industries, the reduction of plastic production will promote the transformation and upgrading of the plastic industry to develop novel eco-friendly plastics or plastic alternatives (degradable and other pollution-free plastics will not be restricted by the growth model), and meet the rigid demand of all aspects of society. Our models predict that in the next 10 years, the plastics industry will gradually return to its usual rate of growth in every aspect of plastic production, and that the cost of disposing of each unit of plastic waste will fall considerably, given technological advances and reduced policy constraints. Therefore, it is a method that sacrifices part of the contemporary interests but is beneficial for long-term development in the future.

4.3.2 Environment

If the plastic industry reduces production by nearly 25MT/year according to our model, plastics will still be harmful and unsafe for the environment in the 10 years to reach the equilibrium point due to its existing large base level (at least 300MT), but such negative effect would shrinking and might even be elimination by 2030. Management of plastic waste, as a worldwide problem, would requires a long-term effort, and a sustainable perspective is needed in this long-run endeavor.

5 Task 4: A Solution for Countries of Viability and Equality

Environmental problems are one of the worldwide challenges. However, the causes and consequences of environmental issues are not equal among countries or regions. The previous section has discussed different goals considering varied contexts of countries. In the following section, we further discuss the goal under the context of equality for countries to make a widely accepted policy. The following issues can be concluded in literacy review:

Issue 1. Developed countries export their plastic waste to less developed countries, which is, in fact, a transfer of environmental crisis, making a more significant gap in the environmental condition between developing countries and developed countries. [39][49]

Issue 2. Economic development, technological development, and cultures differ from country to country, resulting in different waste treatment expenses. For example, with the more advanced technology, the amount of plastic that can be handled would increase, which we have discussed in the previous section. Such an advantage should be well noticed and utilized to keep global waste disposal costs lower.[50]

Issue 3. The political interests of countries differ due to the different development status. Therefore, the model and the rules should be dynamically altered.[21]

Issue 4. Some developing countries produce too much waste plastic waste, making it impossible to dispose of plastic waste properly. So in response, each country should be given the task of building plastic waste disposal facilities in proportion to the amount of plastic waste produced. More facilities should be built if more waste is generated.[40]

5.1 Model: Linear Programming (LP) Problem

5.1.1 Assumption

1. The cost of shipping plastic waste from the country i to country j is paid by country i .
2. Countries participating in this set of rules are willing to disclose related information fully
3. The parameters of the following model will be updated by certain personnel.

5.1.2 Multicriterion optimization

We turn the above equity problem into a multicriterion optimization problem.[50-51] Our goal is to take abovementioned 4 issues into account and find optimal values in trade-off for different goals.

Let $trans_{ij}$ denotes the annual amount of plastic waste transported from country i to country j . ($trans_{ii}$ denotes the annual amount of domestic plastic waste handled). a_{ij} denotes the price per unit weight that country i ship the plastic waste to country j . (a_{ii} denote the cost of handling domestic plastic waste)

We can use different nodes to represent countries, and we can create a weighted directed graph of global plastic waste transport, as shown in Figure 7. The transport information be-

tween countries can be stored in the adjacency matrix of the directed graph.

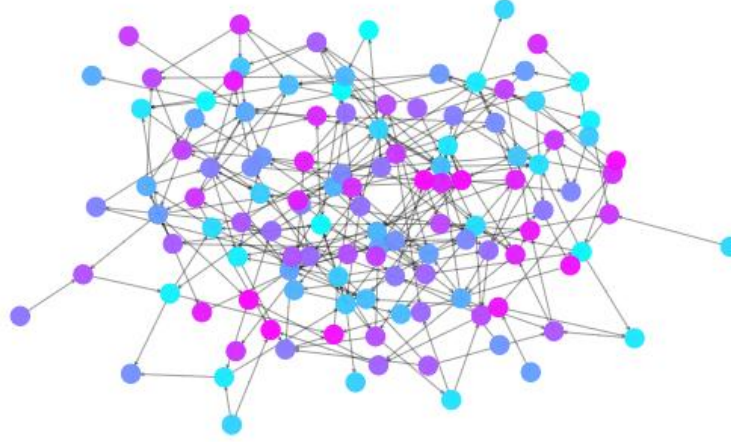


Figure 7: directed graph of plastic waste transportation between countries

We assume the amount of plastic transported as $Trans_{ij}$, the total amount of plastic waste of a country is P_i . The amount that each country can handle is D_i , the expense of each country's handling of the domestic waste is denoted as C_i , n is the total number of countries, and we derive the following multicriterion optimization problem:

$$\text{minimize } \sum_i \sum_j trans_{ij} (i \neq j), C_1, C_2, \dots, C_n \quad (1)$$

$$s.t. \quad \begin{cases} \sum_{j=1}^n trans_{ij} + D_i \geq P_i + \sum_{k=1}^n trans_{ki} & (j \neq i) \quad (2) \\ C_i = \sum_{j=1}^n a_{ij} trans_{ij} - \sum_{k=1}^n a_{ki} trans_{ki} & (k \neq i) \quad (3) \end{cases}$$

The objective function(1) is established that aims to minimize both the amount of waste transported globally and the cost of disposing of plastic waste in each country. This optimization model should be able to reduce the total amount of plastic waste transferred globally, and restricts the transfer of plastic waste from developed countries to less developed countries. Meanwhile, the cost of disposing of plastic waste in each country is considered to minimize the economic burden of each country.

The inequation(2) constrains relevant parameters that ensure waste generated can be managed appropriately in different countries.

The equation(3) is the cost of waste disposal in each country is calculated

After such assumption and operation, this problem can be transformed into a linear programming(LP) problem, which can be solved easily with a predictable result. If this no solution has been found, it indicates that the global plastic waste treatment capacity is not up to the requirements, unable to deal with the global plastic waste. To ensure equity among countries, each country should be given the task of building plastic waste disposal facilities in proportion to the amount of plastic waste generated until the model is solvable. Countries

with more production need to build more processing facilities. Issue 4 can be solved with the abovementioned approach.

5.1.3 Scalarizing Multicriterion Problems and Trade-off Analysis

The set of Pareto optimal values for a multicriterion problem is called the optimal trade-off surface. To find the solution we need in the optimal trade-off surface, We scalarizing multicriterion problems, as shown below.

$$\begin{aligned} & \text{minimize } \sum_i \sum_j \lambda_0 \text{trans}_{ij} + \lambda_1 C_1 + \lambda_2 C_2 + \dots + \lambda_n C_n \\ & \text{s.t. } \begin{cases} \sum_{j=1}^n \text{trans}_{ij} + D_i \geq P_i + \sum_{k=1}^n \text{trans}_{ki} & (j \neq i) \\ C_i = \sum_{j=1}^n a_{ij} \text{trans}_{ij} - \sum_{k=1}^n a_{ki} \text{trans}_{ki} & (k \neq i) \end{cases} \end{aligned}$$

The different λ represent different priorities. The larger the value of λ , the more urgent the problem is and therefore gives a higher priority.

By adjusting λ , we can adjust the priority of different equity issues.

For example, let $\lambda_1 = \lambda_2 = \lambda_3 = \dots = \lambda_n$, which indicates that each country has an equal priority for waste treatment costs.

Let $\sum_1^n \lambda_i = \lambda_0$, which means that the priority of minimizing global plastic waste transshipment is the same as minimizing global waste disposal costs.

$\lambda_1, \lambda_2, \dots, \lambda_n$ can also be adjusted to alter the priorities among different countries. If a country has a natural disaster or a country is economically poor, we can raise those countries' priorities and reduce its burden. This is the representation of humanitarian assistance and equity.

6 Discussion

6.1 Sensitivity Analysis

The six indicators of economy, policy, severity level, industrial scale, regional communication and industrial scale run through Task 1, Task 2 and Task 3. They affect each other as well as the effectiveness of plastic processing methods. For the prediction of their weights, we adopt the quadratic exponential smoothing method, which is highly adaptable and the results are stable. Perturb the 6 main indicators within the 90% confidence bound and then perform curve fitting. We found that the change in the weight of technology and policy has the greatest impact on the model, accounting for 32.1 % and 27.8 % of the weight, respectively, but we also surprisingly found that the linear result change didn't exceed 4.12 % , which roughly changed within [0.0212, 0.0413], reflecting the stability and robustness of our model.

6.2 Stengths and Weaknesses

Stengths

- In order to reach the minimum level of security, we conduct comprehensive and objective **big data analysis** of countries around the world through 6 indicators, and finally divide them into 5 categories with high reliability.
- We convert global security problems into solvable **convex optimization problems**, which is reasonable and effective.
- The **multi-index comprehensive evaluation method** is used to comprehensively measure the main indicators such as economy, policy and regional communication. By means of matrix theory , we can list some logic equations to establish a robust model.
- From the entropy weight method of information theory to the comprehensive decision-making method based on groups, we have optimized the evaluation model of comprehensive indicators and improved the universality to a certain extent.

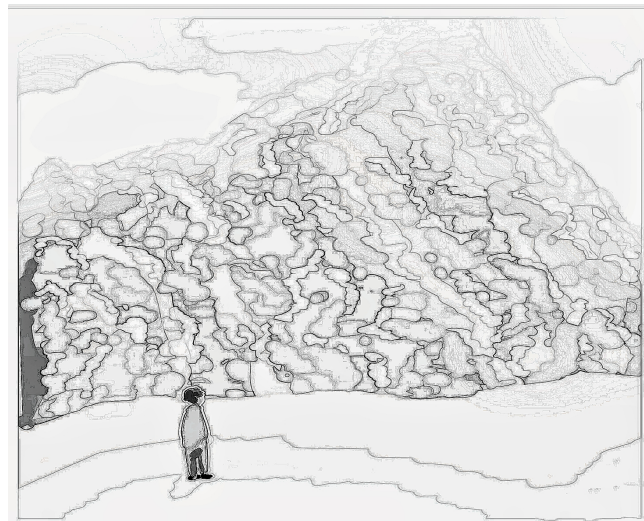
Weaknesses

- In the process of data analysis, we ignored the influence of some small countries, and our model is ideal, which may be resisted by some negative countries.
- It is quite hard for the plastics industry to choose our long-term sustainable development model despite of short-term economic loss
- We used the method of quadratic exponential smoothing to predict the expected weight of influencing factors and established a linear model, while the exponential expectation model may be closer to the real life.

7 Memorandum for ICM

Imagine after days of endless climbing and battling with harsh weathers, you finally reach the top of Mount Everest and just as you were about to admire this sacred land of your childhood dreams, you found yourself struggling amongst a sea of discarded camping gears, used oxygen tanks, ropes, food packaging and countless plastic wastes. You are stunned, disappointed and slightly disheartened, hence after taking a couple of pictures from calculated angles, you decided to travel to somewhere warm to relight your passion for exploring this world.

Perhaps Hawaii's Forbidden Island Niihau would be good, you had always wished to travel there. Following some complicated paper works, you eventually arrived on a helicopter. You reached the beach and got ready to go for a deep dive, yet your tour guide informed you that this pristine beach is closed for the day since it was overwhelmed with plastic detritus. Much of the debris is being washed up from a giant convergence of waste, dubbed the Great Pacific Garbage Patch, swirling around in the Pacific Ocean between California and Hawaii.



One of the biggest the global emergencies humans are facing today is the over exploitation of plastic usage. From filthy beaches and stinking cities to plastic-choked waterways, plastic waste is threatening many of Earth's most beautiful destinations and last remaining wildernesses. Despite the urgent situation, many are still turning a blind eye upon such matters, however, many activists are fighting to save our world through organisations and personal actions.

Immediate actions must be taken, and a more effective approach is through the intervalence of the policymakers. A target can be set to realize the goal of eliminating plastic-induced environmental problems.

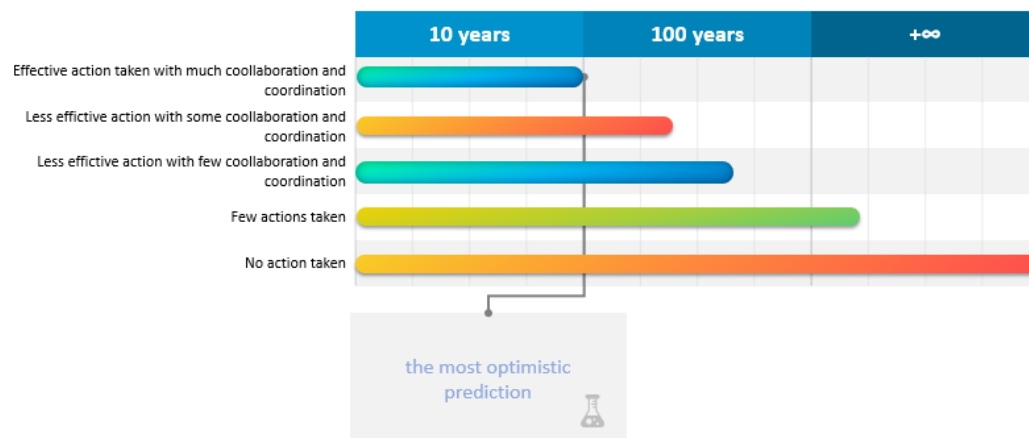
According to the model we established, we estimated the Total Plastic Production under Annual Maximum Capacity of plastic waste as 213 Million Tons, while the predicted current level of plastic production is 437MT. The gap between the safety capacity and the current level indicates that at least 48.74% of production should be eliminated. To make this happen, the

target can be distributed to countries based on their circumstances and use the growth rate of the plastic industry as the controlling variable.

Controlling the growth rate of the plastic industry is controlling the source of plastic production, and different target rate is set for different countries to achieve the overall goal. Our analysis ensures that such goals are realistic for the corresponding countries. Therefore, if countries successfully adapted and achieved the goal, or even overachieved the target, the overall process of beating plastic-induced environmental problem will be accelerated. It can be optimistically estimated that it is viable for our world to reach the ecological -safe level for plastics in just ten years if efforts are made, and individual goals are achieved. However, hinderance will be put in achieving the target if individual targets are not met in reality.

Moreover, such the act requires collaboration and coordination among countries, and proper collaboration will also foster the process of dealing with the environmental problem.

Therefore, the effort each country makes, as well as the collaboration and coordination, are essential factors that determine whether our human beings can successfully beat the plastic crisis we are facing today. A timeline is illustrated in the figure below,



There is still a lot more for us to do and take responsibility of, since this world's future is dependent on your decisions now, the choices you make today, whether it is to refuse to use a plastic straw or to investigate further into the replacement of plastic products. The effects of the overuse of plastic will affect every country, so it's time to unite together and overcome this challenge together!

References

- [1] Strom, E. T., & Rasmussen, S. C. (Eds.). (2011). 100+ years of plastics. Leo Baekeland and Beyond. American Chemical Society.
- [2] Reboul, P. (1998). Pioneers of plastics. *Interdisciplinary Science Reviews*, 23(2), 169-173.
- [3] Lithner, D., Larsson, Å., & Dave, G. (2011). Environmental and health hazard ranking and assessment of plastic polymers based on chemical composition. *Science of the Total Environment*, 409(18), 3309-3324.
- [4] Thompson, R. C., Swan, S. H., Moore, C. J., & vom Saal, F. S. (2009). Our plastic age. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 364(1526), 1973–1976.
- [5] Geyer, R., Jambeck, J. R., & Law, K. L. (2017). Production, use, and fate of all plastics ever made. *Science Advances*, 3(7), e1700782.
- [6] Kershaw, P., Katsuhiko, S., Lee, S., & Woodring, D. (2011). Plastic debris in the ocean. United Nations Environment Programme.
- [7] Verma, R., Vinoda, K. S., Papireddy, M., & Gowda, A. N. S. (2016). Toxic pollutants from plastic waste-a review. *Procedia Environmental Sciences*, 35, 701-708.
- [8] Mudgal, S., Lyons, L., Bain, J. et al. (2011) Plastic Waste in the Environment – Revised Final Report for European Commission DG Environment. Bio Intelligence Service. <http://www.ec.europa.eu/environment/waste/studies/pdf/plastics.pdf>
- [9] Allsopp, M., Walters, A., Santillo, D. & Johnston, P. (2006) Plastic Debris in the World's Oceans. Greenpeace International.
- [10] Ragaert, K., Delva, L., & Van Geem, K. (2017). Mechanical and chemical recycling of solid plastic waste. *Waste Management*, 69, 24-58.
- [11] Santos, A. S. F., Teixeira, B. A. N., Agnelli, J. A. M., & Manrich, S. (2005). Characterization of effluents through a typical plastic recycling process: An evaluation of cleaning performance and environmental pollution. *Resources, conservation and recycling*, 45(2), 159-171.
- [12] Shen, L., & Worrell, E. (2014). Plastic recycling. In *Handbook of Recycling* (pp. 179-190). Elsevier
- [13] Meidiana, C., & Gamse, T. (2010). Development of waste management practices in Indonesia. *European journal of scientific research*, 40(2), 199-210.
- [14] Hoornweg, D., & Bhada-Tata, P. (2012). What a waste: a global review of solid waste management.

- [15] Kgosiesele, E., & Zhaohui, L. (2010). An evaluation of waste management in Botswana: Achievements and challenges. *Journal of American Sciences*, 6, 144-150.
- [16] Agamuthu, P., Fauziah, S. H., Khidzir, K. M., & Aiza, A. N. (2007, September). Sustainable waste management-Asian perspectives. In *Proceedings of the international conference on sustainable solid waste management* (Vol. 5, p. 15).
- [17] Dhokhikah, Y., & Trihadiningrum, Y. (2012). Solid waste management in Asian developing countries: challenges and opportunities. *Journal of Applied Environmental and Biological Sciences*, 2(7), 329-335.
- [18] Karani, P., & Jewasikiewitz, S. M. (2007). Waste management and sustainable development in South Africa. *Environment, Development and Sustainability*, 9(2), 163-185.
- [19] Imam, A., Mohammed, B., Wilson, D. C., & Cheeseman, C. R. (2008). Solid waste management in Abuja, Nigeria. *Waste management*, 28(2), 468-472.
- [20] Kolikkathara, N., Feng, H., & Stern, E. (2009). A purview of waste management evolution: Special emphasis on USA. *Waste management*, 29(2), 974-985.
- [21] The United Nation. (2018). *Legal Limits on Single-Use Plastics and Microplastics: A Global Review of National Laws and Regulations*
- [22] Gourmelon, G. (2015). Global plastic production rises, recycling lags. New Worldwatch Institute analysis explores trends in global plastic consumption and recycling. *Recupero de*
- [23] Barnes, D. K., Galgani, F., Thompson, R. C., & Barlaz, M. (2009). Accumulation and fragmentation of plastic debris in global environments. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 364(1526), 1985-1998.
- [24] Zheng, Y., Yanful, E. K., & Bassi, A. S. (2005). A review of plastic waste biodegradation. *Critical reviews in biotechnology*, 25(4), 243-250.
- [25] Gautam, R., Bassi, A. S., & Yanful, E. K. (2007). A review of biodegradation of synthetic plastic and foams. *Applied biochemistry and biotechnology*, 141(1), 85-108
- [26] Suhov, Y., Stuhl, I., Sekeh, S. Y., & Kelbert, M. (2016). Basic inequalities for weighted entropies. *Aequationes mathematicae*, 90(4), 817-848.
- [27] Qiyue, C. (2010). Structure entropy weight method to confirm the weight of evaluating index. *Systems Engineering Theory & Practice*, 30(7), 1225-1228.
- [28] Ven, A. H. V. D., & Delbecq, A. L. (1974). The effectiveness of nominal, Delphi, and interacting group decision making processes. *Academy of management Journal*, 17(4), 605-621.
- [29] Chen, Z., & Yang, W. (2011). A new multiple attribute group decision making method in intuitionistic fuzzy setting. *Applied Mathematical Modelling*, 35(9), 4424-4437.

- [30] Alvaredo, F., Chancel, L., Piketty, T., Saez, E., & Zucman, G. (Eds.). (2018). World inequality report 2018. Belknap Press.
- [31] Lluís Carrion-i-Silvestre, J., Del Barrio-Castro, T., & López-Bazo, E. (2005). Breaking the panels: an application to the GDP per capita. *The Econometrics Journal*, 8(2), 159-175.
- [32] Ayal, E. B., & Karras, G. (1998). Components of economic freedom and growth: An empirical study. *The Journal of Developing Areas*, 32(3), 327-338.
- [33] Boulhol, H., A. de Serres and M. Molnar (2008), "The Contribution of Economic Geography to GDP Per Capita", OECD Economics Department Working Papers, No. 602, OECD Publishing, Paris,
- [34] Xanthos, D., & Walker, T. R. (2017). International policies to reduce plastic marine pollution from single-use plastics (plastic bags and microbeads): a review. *Marine pollution bulletin*, 118(1-2), 17-26.
- [35] Nielsen, T. D., Holmberg, K., & Strippel, J. (2019). Need a bag? A review of public policies on plastic carrier bags—Where, how and to what effect?. *Waste management*, 87, 428-440.
- [36] Kibria, G. (2017). Plastic Waste, Plastic Pollution—A Threat to All Nations. Technical Report 2017.
- [37] Tietenberg, T. H. (1990). Economic instruments for environmental regulation. *Oxford Review of Economic Policy*, 6(1), 17-33.
- [38] Helm, D. (2005). Economic instruments and environmental policy. *Economic and social review*, 36(3).
- [39] UN Comtrade Database <https://comtrade.un.org/data/>
- [40] D. Hoornweg, P. Bhada-Tata, "What a waste: A global review of solid waste management" (The World Bank, 2012); <https://openknowledge.worldbank.org/handle/10986/17388>
- [41] Harper, C. A. (2000). *Modern plastics handbook*. McGraw Hill Professional.
- [42] Pearson, K. (1901). LIII. On lines and planes of closest fit to systems of points in space. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 2(11), 559-572.
- [43] MacQueen, J. (1967, June). Some methods for classification and analysis of multivariate observations. In *Proceedings of the fifth Berkeley symposium on mathematical statistics and probability* (Vol. 1, No. 14, pp. 281-297).
- [44] Tan, K. M., & Witten, D. (2015). Statistical properties of convex clustering. *Electronic journal of statistics*, 9(2), 2324.

- [45] Cheeger, J. (1969). A lower bound for the smallest eigenvalue of the Laplacian. In Proceedings of the Princeton conference in honor of Professor S. Bochner (pp. 195-199).
- [46] Kimmel, R. (2019). Processing, Analyzing and Learning of Images, Shapes, and Forms (Vol. 20). North Holland.
- [47] D. Hoornweg, P. Lam, M. Chaudhry, "Waste management in China: Issues and recommendations" (The World Bank Urban Development Working Papers no. 9, 2005).
- [48] Filatov, V. V., Zaitseva, N. A., Larionova, A. A., Zhenzhebir, V. N., Polozhentseva, I. V., Takhumova, O. V., & Kolosova, G. M. (2018). State Management of Plastic Production Based on the Implementation of UN Decisions on Environmental Protection. *Ekoloji Dergisi*, (106).
- [49] Brooks, A. L., Wang, S., & Jambeck, J. R. (2018). The Chinese import ban and its impact on global plastic waste trade. *Science advances*, 4(6), eaat0131.
- [50] Stadler, W. (Ed.). (1988). *Multicriteria Optimization in Engineering and in the Sciences* (Vol. 37). Springer Science & Business Media.
- [51] Geoffrion, A. M., Dyer, J. S., & Feinberg, A. (1972). An interactive approach for multicriterion optimization, with an application to the operation of an academic department. *Management science*, 19(4-part-1), 357-368.

Supplemental Material: The result of clustering

Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
Qatar	Netherlands	Russia	Monaco	Germany
Kuwait	Finland	China	Norway	Belgium
Puerto Rico	Israel	Brazil	Ireland	Japan
Malta	France	South Africa	Iceland	
Cyprus	Italy	Indonesia	Qatar	
Turks and Cacos Islands	Spain	Philippines	Singapore	
Slovenia	Chile	Vietnam	United States	
Bahrain	Panama	Nigeria	Denmark	
Portugal	Romania	India	Australia	
Saudi Arabia	Argentina	Pakistan	Sweden	
Eritrea	Turkey	Myanmar	Canada	
Northern Mariana Islands	Colombia	Sudan	United Arab Emirates	
Curacao	Albania	Somalia	United Kingdom	
Lithuania	Guyana		New Zealand	
Barbados	Tunisia		Greece	
Latvia	Morocco			
Uruguay	Papua New Guinea			
Trinidad and Tobago	Kenya			
Antigua and Barbuda	Bangladesh			
Seychelles	Cameroon			
Oman	Senegal			
Palau	Mauritania			
Poland	Benin			
Croatia	Haiti			
Costa Rica	Mozambique			
Malaysia				
Mauritius				
Grenada				
Maldives				
Equatorial Guinea				
Nauru				
Mexico				
Bulgaria				
Montenegro				
Cuba				
Lebanon				
Dominican Republic				
Gabon				
Dominica				
Thailand				
Libya				
Peru				
Ecuador				
Fiji				
Suriname				
Bosnia and Herzegovina				
Namibia				
Iraq				
Jamaica				
Belize				
Georgia				
Guatemala				
Tonga				
Jordan				
Samoa				
Algeria				
Sri Lanka				
El Salvador				
Marshall Islands				
Tuvalu				
Angola				
Vanuatu				
Ukraine				
Djibouti				
Honduras				
Ghana				
Solomon Islands				
Nicaragua				
Sao Tome and Principe				
Kiribati				
Cambodia				
Comoros				
Tanzania				
Guinea				
Guinea-Bissau				
Togo				
Liberia				
Sierra Leone				
Madagascar				